

Option B – Medical Physics

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)	Heater to produce / boil off [free] electrons / thermionic emission (1) accept heater gives energy to electrons Avoid collisions [on way to target] (1)	2			2		
		(ii)	$\frac{100 \times 10^{-3}}{1.6 \times 10^{-19}}$ (1) = 6.25×10^{17} or 6.3×10^{17} with no units (1) accept electrons per second		2		2	2	
		(iii)	Application of conservation of energy: $\frac{1}{2} mv^2 = eV$ (1) Rearrangement: $v^2 = \frac{2eV}{m}$ (1) $v = 8.3$ or 8.4×10^7 [m s ⁻¹] (1)		1 1 1		3	3	
		(iv)	Use of $eV = \frac{hc}{\lambda_{\min}}$ (1) Rearrangement: $\lambda_{\min} = \frac{hc}{20000e}$ (1) $\lambda_{\min} = 6.2 \times 10^{-11}$ [m] (1)	1	1 1		3	3	
	(b)	(i)	Alternating voltage used (1) accept alternating current used Causes piezoelectric crystal to vibrate (1) Ultrasound reflected and Doppler shift detected (1)	3			3		

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		(ii)	$v = \frac{\Delta f c}{2 f_0 \cos 37}$ (1) Rearrangement and substitution $v = 0.11[3]$ [m s ⁻¹] (1)		2		2	2	
	(c)		Any 5 × (1) from: • MRI and ultrasound and CT – good images [only] (1) • X-ray – poor image / soft tissue contrast (1) • Radioactive tracer best or gives more information (1) • Tracer goes to hyperactive [parts of] thyroid (1) • Imaged using gamma camera (1) • Any further detail e.g. Initial diagnosis using blood (check levels of thyroid hormones) Iodine gives added contrast (for CT scan) Tracer can locate problem areas Tracer can be used to “kill” overactive thyroid / thyroid cancer			5	5		
			Question 8 total	6	9	5	20	10	0